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SANTOSH, TANGAIL-1902



DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Lab Report

Lab Report No : 02

Lab report name : Implementation of CSS Inclusion Methods and CSS Selectors in Web Design

Course Title : Web Application Development Lab

Course Code : ICT-3204

Submitted By	Submitted To
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1. Lab Title: Implementation of CSS Inclusion Methods and CSS Selectors in Web Design.

2. Lab No: 02

3. Introduction:

Cascading Style Sheets, fondly referred to as CSS, is a simply designed language intended to simplify the process of making web pages presentable. It controls color, font, spacing, layout and overall looking style of html element. CSS can be added in three ways:

1. Inline CSS
2. Internal CSS
3. External CSS

CSS selectors are used to target html elements for styling. Here we discuss six types of selectors:

1. Element selector
2. Class selector
3. ID selector
4. Universal selector
5. Group selector
6. Hover selector

4. Objectives:

1. To learn different CSS inclusion method.
2. To understand CSS selectors work.
3. To design a simple styled webpage use CSS

4. CSS inclusion methods:

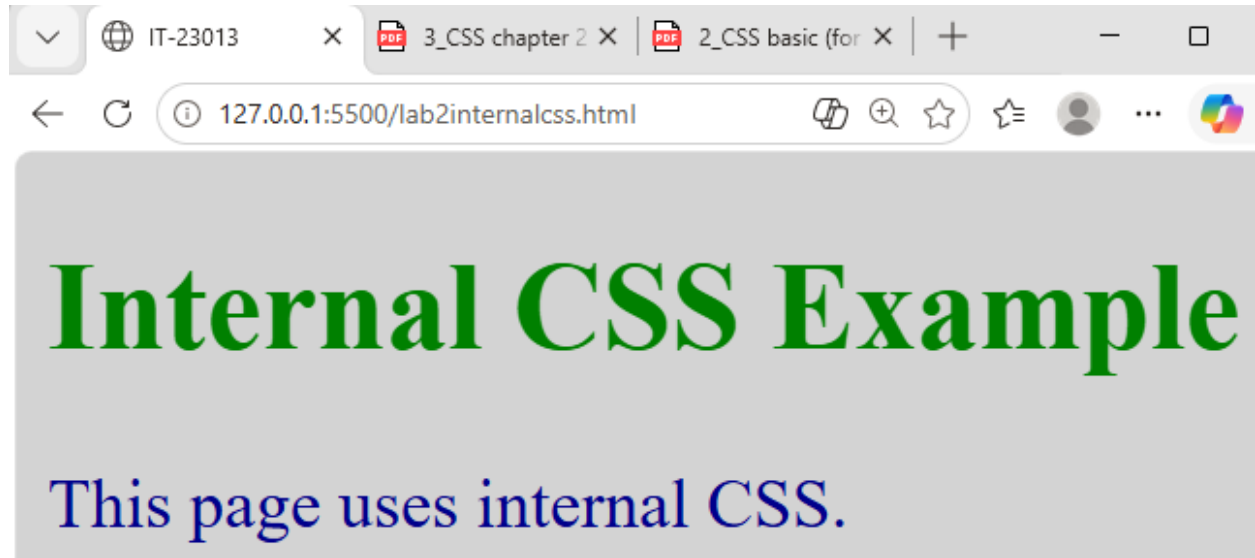
4.1: Inline css:

CSS is written directly inside HTML tags using the style attribute. It is suitable for small styling. It is not recommended use always.



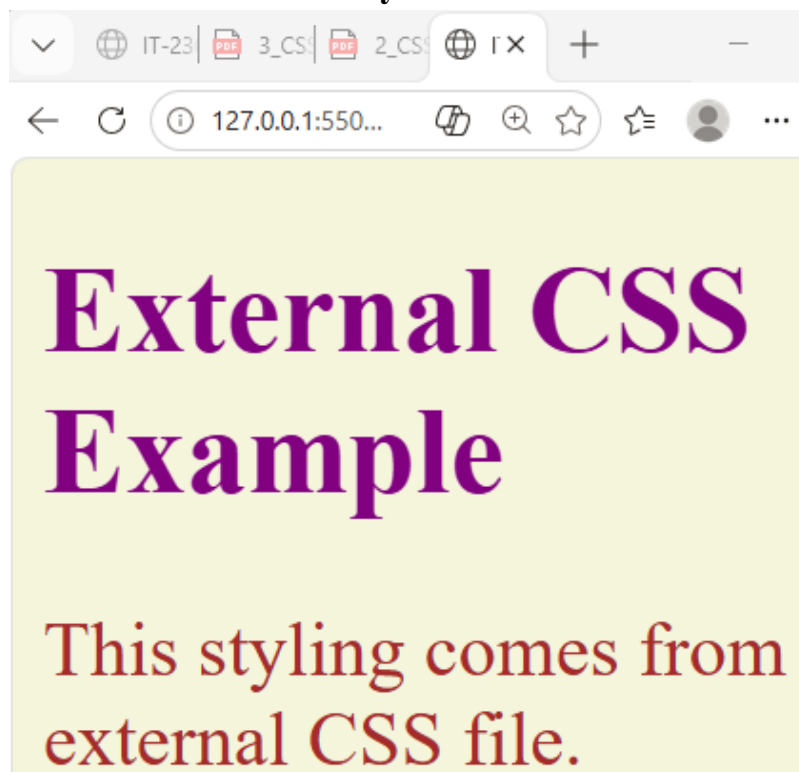
4.2: Internal css:

CSS is written inside HTML tags using the style tag within a single document. It is suitable for simple webpage styling. It is not recommended use always.



4.3: External css:

CSS is written in separate .css file. It is the best method for large websites. It is recommended use always.

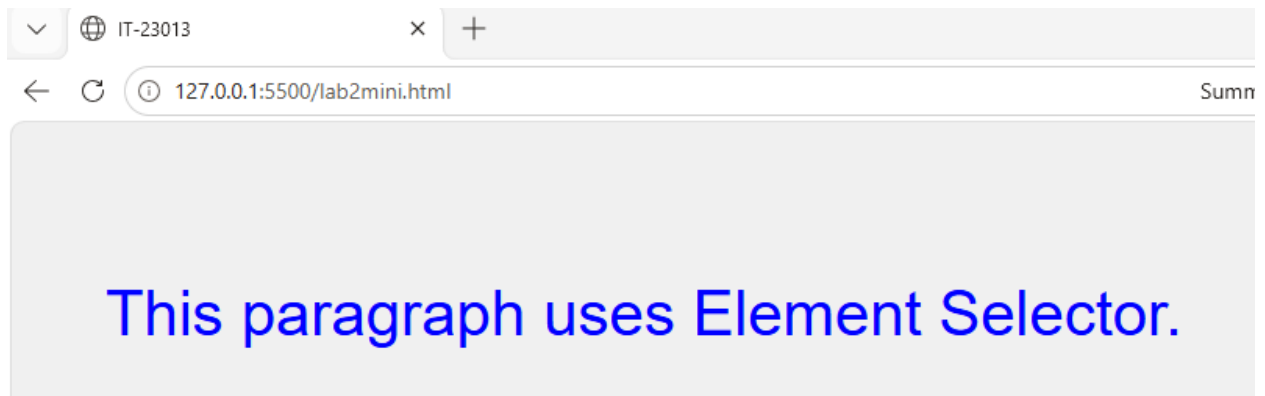


5.CSS Selectors:

5.1: Element Selector:

```
<style>
p{
  color: blue;
}
</style>
```

```
<p>Hello World</p>
```

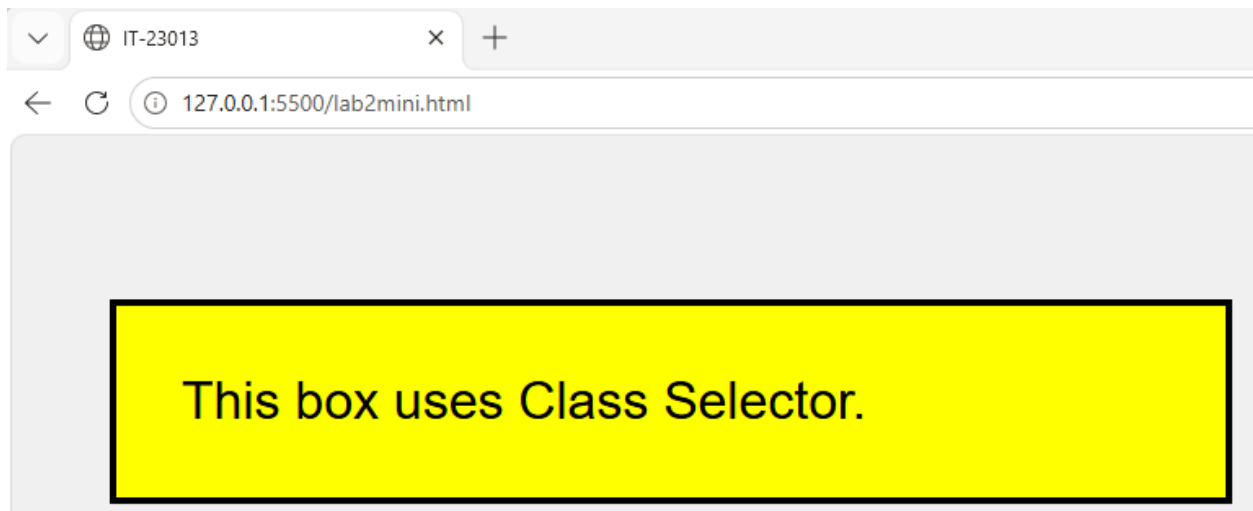


Here target all <p> elements.

5.2: Class Selector:

```
<style>
.box{
  background-color: yellow;
}
</style>
```

```
<div class="box">
  Class Selector Example
</div>
```

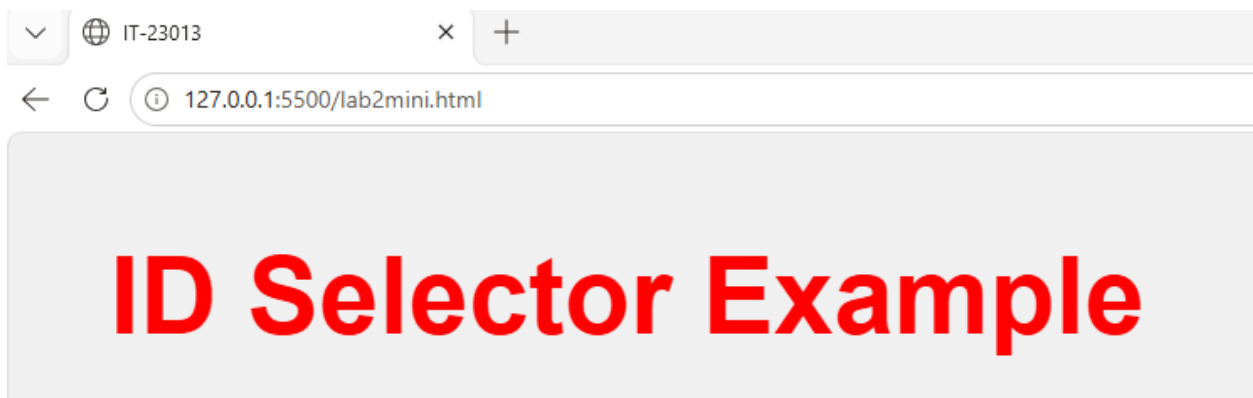


Here targets elements use class name.

5.3: ID Selector:

```
<style>
#title{
  color: red;
}
</style>
```

```
<h1 id="title">ID Selector</h1>
```



Target a unique element use id.

5.4: Universal Selector:

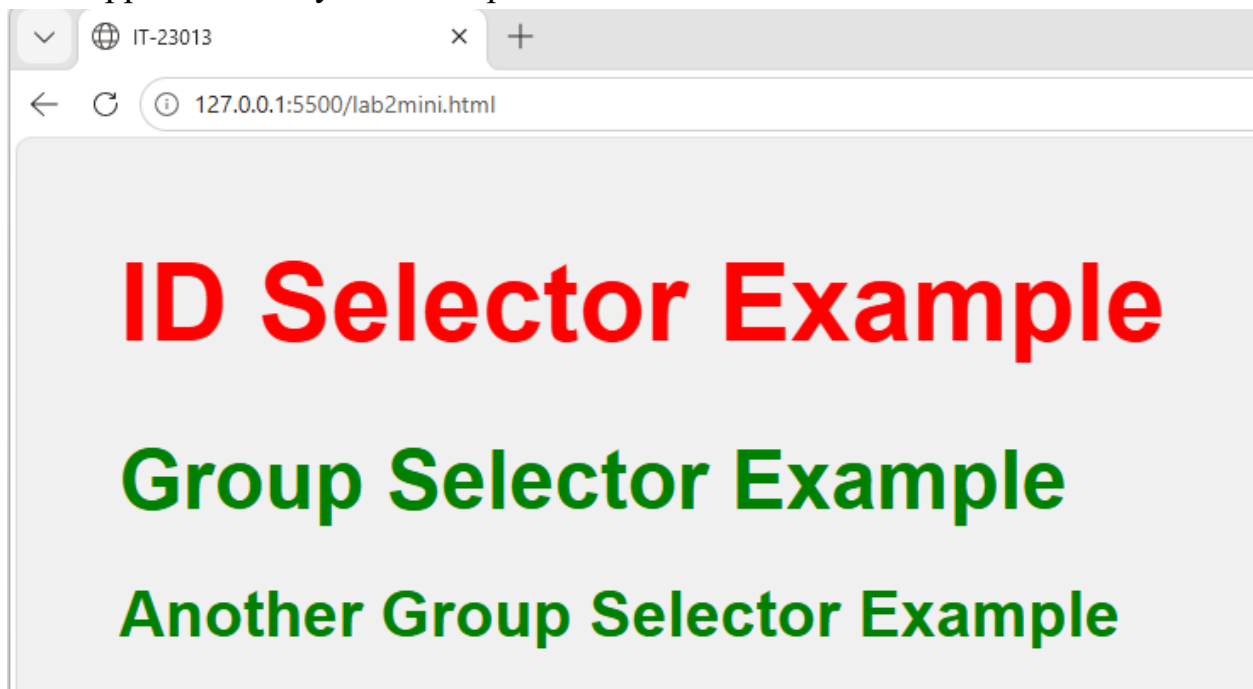
```
<style>
*{
  font-family: Arial;
}
</style>
```

Here applies style to all element.

5.5: Group Selector:

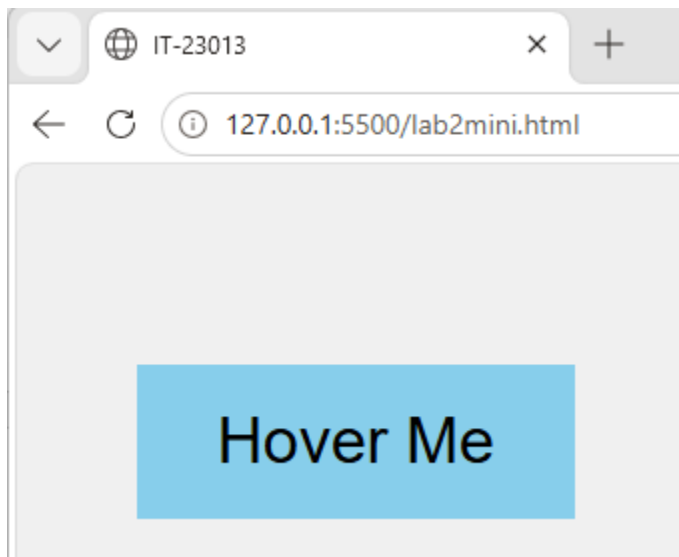
```
<style>
h1, p{
  color: green;
}
</style>
```

Here applies same style to multiple element.

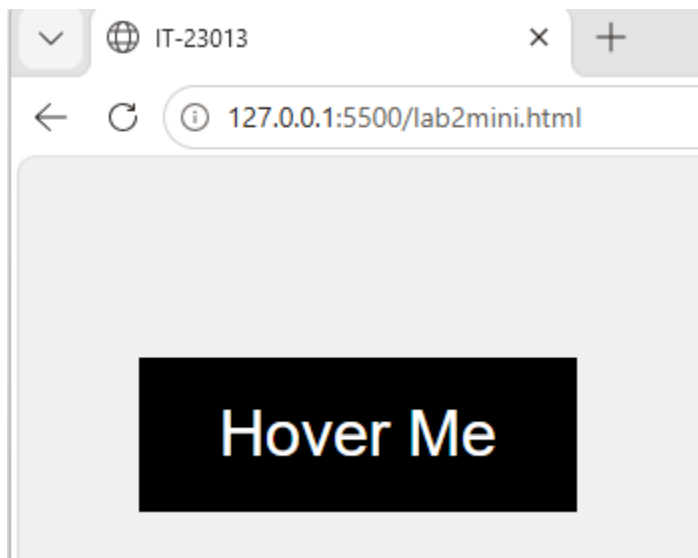


5.5: Hover Selector:

```
<style>  
button:hover{  
    background-color: black;  
    color: white;  
}  
</style>  
<button>Click Me</button>
```



Normal



After mouse hovers over element

Here change style when mouse hovers over element.

7. Conclusion:

In this lab, successfully implemented different CSS inclusion methods and CSS selectors in web design. I learn how CSS can be applied to html document using inline css, internal css and external css. I also learn how work different selectors such as Element, Class, ID, Universal, Group, and Hover selectors to style webpage elements effectively. This lab help us to gain practice knowledge about webpage styling, improving and understanding of basic web design concept using CSS.

Source Code:

4.1:

```
<!DOCTYPE html>
<html>
<head>
  <title>IT-23013</title>
</head>
<body>

<h1 style="color: blue;">Welcome to CSS</h1>

<p style="font-size:20px; color:red;">
  This is inline CSS example.
</p>

</body>
</html>
```

4.2:

```
<!DOCTYPE html>
<html>
<head>
  <title>IT-23013</title>

  <style>
```



```

    body{
        background-color: lightgray;
    }

    h1{
        color: green;
    }

    p{
        font-size: 18px;
        color: darkblue;
    }
</style>

</head>
<body>

<h1>Internal CSS Example</h1>

<p>This page uses internal CSS.</p>

</body>
</html>

```

4.3: HTML file:

```

<!DOCTYPE html>
<html>
<head>
    <title>IT-23013</title>

    <link rel="stylesheet" href="lab2external.css">

</head>
<body>

<h1>External CSS Example</h1>

<p>This styling comes from external CSS file.</p>

</body>
</html>

```

CSS file:

```
body{
  background-color: beige;
}

h1{
  color: purple;
}

p{
  font-size: 20px;
  color: brown;
}
```

5: The following code use for the output of all selectors:

HTML code:

```
<!DOCTYPE html>
<html>

<head>
  <title>IT-23013</title>

  <!-- External CSS Link -->
  <link rel="stylesheet" href="lab2mini.css">
</head>

<body>

  <!-- Hover Selector -->
  <button>
    Hover Me
  </button>
  <br>
  <!-- Element Selector -->
  <p>
    This paragraph uses Element Selector.
  </p>
  <!-- Class Selector -->
  <div class="box">
    This box uses Class Selector.
```

```
</div>

<!-- ID Selector -->
<h1 id="title">
  ID Selector Example
</h1>

<!-- Group Selector -->
<h2>
  Group Selector Example
</h2>

<h3>
  Another Group Selector Example
</h3>
<br>

</body>

</html>
```

CSS file:

```
/* Universal Selector */
*{
  font-family: Arial;
  margin: 0;
  padding: 0;
}

/* Body Styling */
body{
  background-color: #f0f0f0;
  padding: 30px;
}

/* ID Selector */
#title{
  color: red;
  margin-bottom: 20px;
}

/* Group Selector */
h2, h3{
  color: green;
  margin-top: 15px;
```

```
}

/* Element Selector */
p{
  color: blue;
  font-size: 20px;
  margin-top: 20px;
}

/* Class Selector */
.box{
  background-color: yellow;
  padding: 20px;
  width: 300px;
  margin-top: 20px;
  border: 2px solid black;
}

/* Button Design */
button{
  padding: 10px 20px;
  background-color: skyblue;
  border: none;
  font-size: 16px;
  cursor: pointer;
  margin-top: 20px;
}

/* Hover Selector */
button:hover{
  background-color: black;
  color: white;
}
```